



CS1 exercise session

About the exercise

- ▷ Slides from Joel Gmür
- ▷ Intuitive understanding
- ▷ Visualisations will be added
- ▷ All content on website

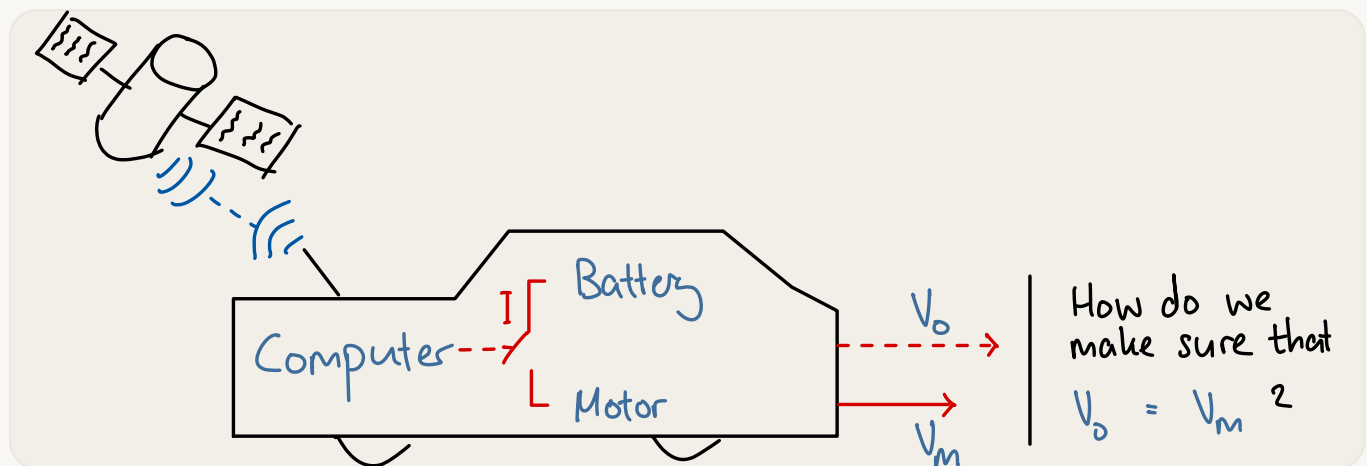
About the lecture

- ▷ 2-page summary sheet given — 40 pages handwritten summary on top
- ▷ Lecture notes PDF
- ▷ Study Center starts 3rd week



What is control?

Let's build intuition with an example.



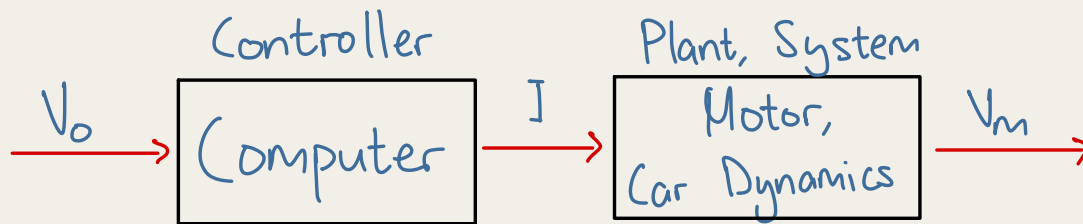
We command the car's computer to move the car with **speed** V_o . But the computer can only control the **current** I going into the motor — it knows the car's true speed through **measurement** V_m of the satellite.

How does the computer know how much current it should send?

— this is the central question of control.

To solve this we will first need to learn many techniques. For you to keep an overview, you must keep the greater picture in mind.

So let's abstract the car example (*for now without the satellite*).



In the sketch, V_0 , I , V_m are called **signals**, while the computer and the motor are called **systems**.

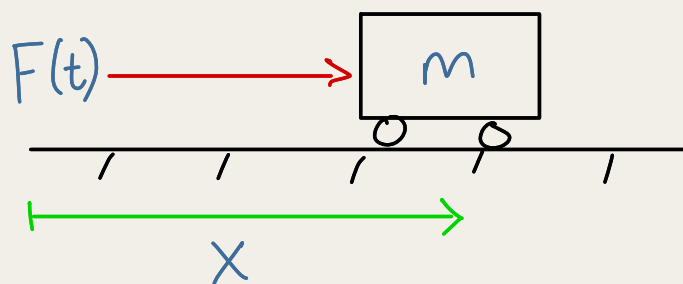
A system **manipulates** a signal.

What are signals and systems?

Signal — a function of time (or frequency).

System — any model that possibly has inputs and/or outputs.

Example:



mass m
force $F(t)$
position x

System — the car with mass m

Signals — $F(t)$ (input), $X(t)$ (output)

We can model systems with differential equations.

Here:

$$F(t) = m\ddot{X}(t) \Rightarrow \ddot{X}(t) = \frac{1}{m} \sin(t)$$

$$\dot{X}(t) = -\frac{1}{m} \cos(t) + C_1$$

$$X(t) = -\frac{1}{m} \sin(t) + C_1 t + C_2$$

From initial conditions: $C_1 = \frac{1}{m}, C_2 = 0$

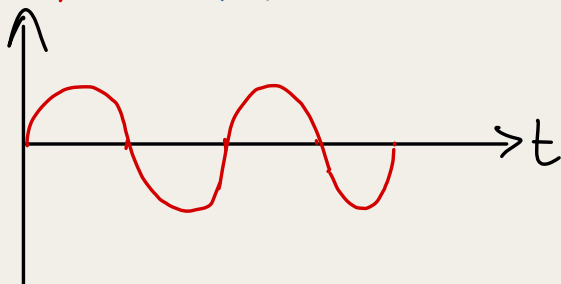
$$X(t) = \frac{1}{m} (-F(t) + t)$$

Assume

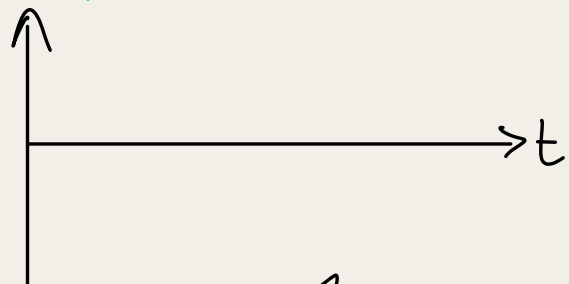
$$X_0 = 0, V_0 = 0$$

In the end, the system is the underlying equation **structure** and the **parameters**. It defines how the input signal is manipulated.

$$F(t) = \sin(t)$$

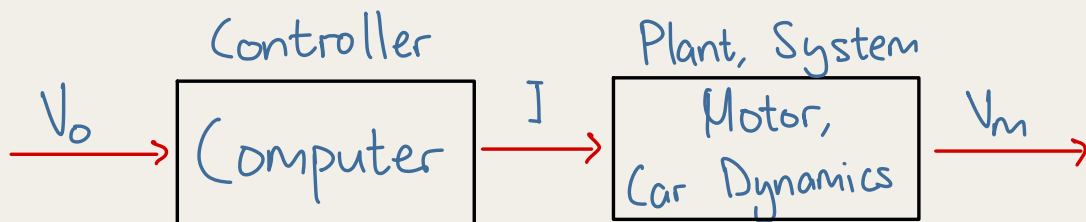
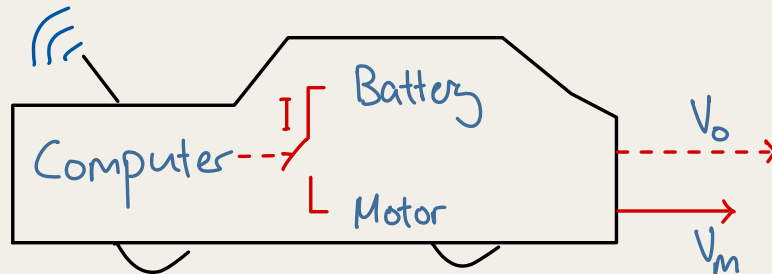


$$x(t) = ?$$



System

Wait, why did we just look at this?

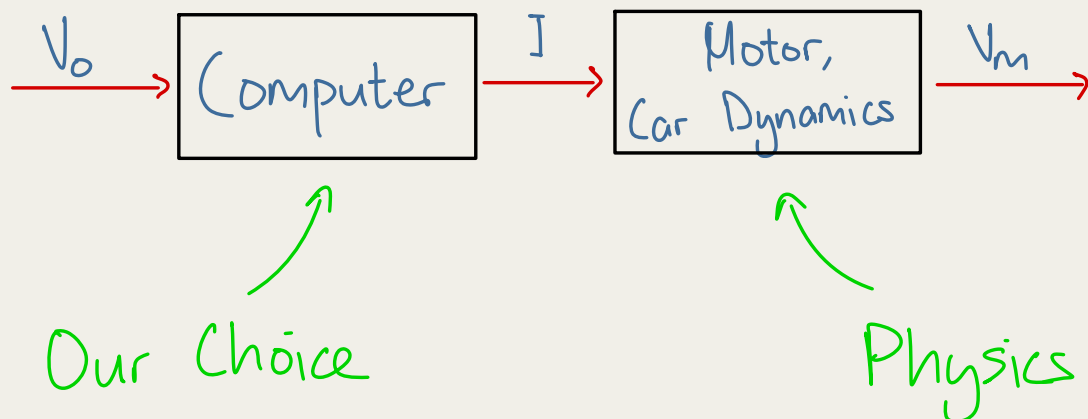


Blocks are systems → Arrows are signals

To figure out how to **control** the car, we need to know the diff' eqns that describe the behaviour of the blocks — this is why the first part of the course will be **modeling**.

If every system is just defined by physics,
then what even is our job as engineers?

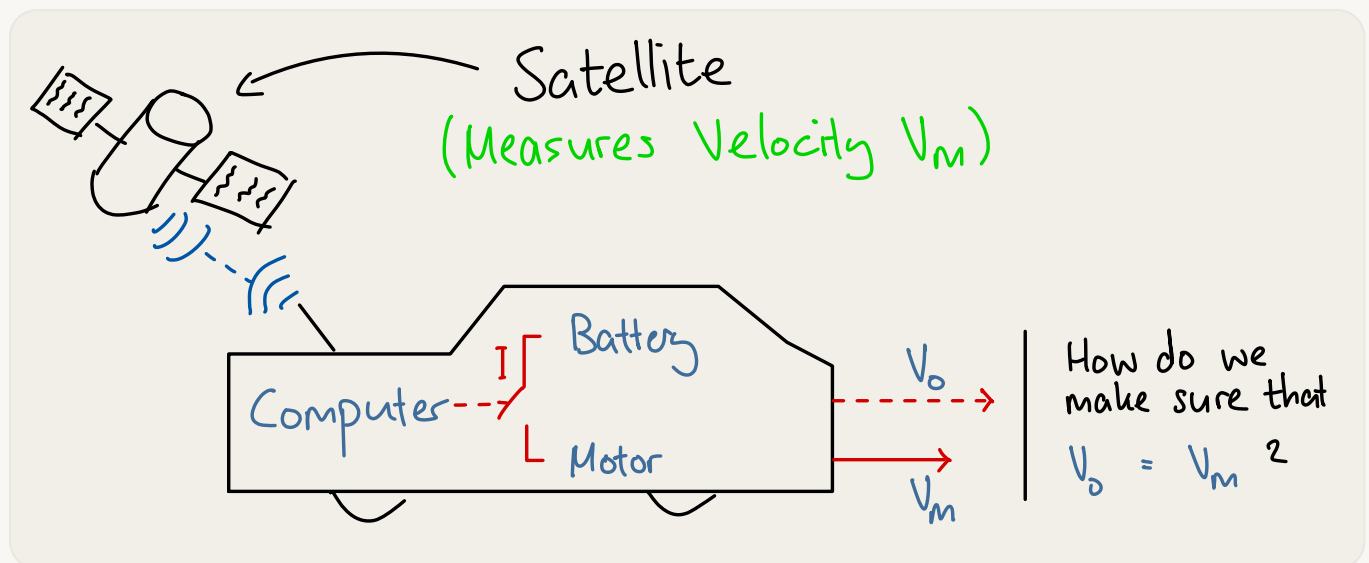
— *this is not true!!*



Some blocks are predefined. The ones that we can design (**computer**), are called **controllers**. We need to design controllers such that the **total system** does what we want. This is hard because of:

Model uncertainties, external disturbances, inaccurate sensors...

These problems will be addressed by introducing **feedback**. Back to our example:



Think about how the satellite could improve this problem...

All of what I showed you was massively simplified and not covered yet. (you are not expected to understand it now) — this was just so you have the big picture in mind.

Intermediate Quiz

Q1) The Input is what we tell the computer to do. True / **False**

We give the computer the reference — it calculates what input the motor should receive.

Q2) Each block represents a physical system. True / **False**

The controller is a software program.

Q3) Not all control systems are electrical. **True** / False

Example: the Watt governor.